ABC AUTOMOTIVE COMPONENTS PVT. LTD.

Predictive Maintenance & Machine Health Intelligence System

REPORT DATE: 2026-10-07 17:33
PLANT LOC: Plant-01 (Precision Division)
AUDIT STATUS: AI Verified

MACHINE HEALTH REPORT CARD: CNC-004 - CNC MACHINE #4

CURRENT HEALTH SCORE	FAILURE PROBABILITY (7D)	RISK LEVEL	MAINTENANCE PRIORITY
42.0 / 100	87.2%	CRITICAL	P1

1. Asset Specifications & Operating Parameters

Machine ID:	CNC-004	**Machine Type:**	CNC Machine
Manufacturer:	Fanuc	**Model Number:**	VCN-530C
Department:	Precision Machining	**Production Line:**	Transmission Gear Line
Asset Age:	1.5 Years	**Warranty Status:**	Active
Operating Hours:	5,503 hrs	**Current Utilization:**	91.7%
Shift Pattern:	2 Shifts (16h)	**Criticality Level:**	Critical

2. Real-Time Telemetry & Deviation Analysis

Sensor Parameter	Current Value	Nominal Range	Baseline Deviation	Telemetry Status
Temp	88.4 °C	50 - 72 °C	+44.9%	Critical Anomaly
Rpm	4850.0 RPM	2500 - 5200 RPM	+26.0%	Normal
Vibration	7.85 mm/s	1.0 - 3.8 mm/s	+227.1%	Critical Anomaly
Pressure	6.11 bar	4.5 - 6.8 bar	+8.1%	Normal
Current	37.5 A	12 - 28 A	+87.5%	Critical Anomaly
Voltage	424.4 V	395 - 415 V	+4.8%	Critical Anomaly
Power	15.65 kW	8 - 22 kW	+4.3%	Normal
Oil Level	31.8 %	65 - 95 %	-60.2%	Critical Anomaly
Humidity	54.56 %	40 - 60 %	+9.1%	Normal
Load	94.0 %	55 - 80 %	+39.3%	Critical Anomaly

3. AI Diagnostics & Risk Factor Attribution

Predicted Failure Mode: Bearing / Spindle Failure (Diagnostic Confidence: 93.3%)

- **Factor 1:** Elevated Vibration (+28.0% Risk Contribution) [Mechanical]
- **Factor 2:** High Operating Temperature (+22.4% Risk Contribution) [Thermal]

• **Factor 3:** Electrical Current Surge (+18.0% Risk Contribution) [Electrical]

4. Recommended Maintenance Protocol

Urgency / Window:	**CRITICAL (P1)** — Within 24 hours (Immediate Intervention)
Recommended Action:	Immediate shutdown or load de-rating. Dispatch lead technician for comprehensive Bearing / Spindle Failure inspection, bearing ultrasound check, and lubricant replacement.
Diagnostic Rationale:	Telemetry displays severe compound anomalies. Failure probability has surged to 87.2%, with high likelihood of catastrophic breakdown during current shift.
Consequence If Ignored:	Unplanned spindle seizure or motor burnout resulting in 12-18 hours of production stoppage, ■1,80,000+ repair costs, and delivery delays to OEM assembly.
Estimated Production Impact:	Potential Downtime: **18.1 hrs** | Parts at Risk: **1991 units** | Breakdown Exposure: **INR 366,185**